

Question

Write a Java program to display all letter combinations for the entered phone digits.

Steps for Execution

1. Save the code as `LetterCombinations.java`.
2. Compile the program.
3. Run `LetterCombinations`.
4. Enter digits when prompted.

Algorithm

1. Read the digit string.
2. If it is empty, display an empty list.
3. For each digit, find its letters on a phone keypad.
4. Add each letter to the combinations already created.
5. Display all combinations.

UML Class Diagram

```
+-----+
|      LetterCombinations      |
+-----+
| + main(args: String[]): void |
+-----+

    | uses
    +---- Scanner
    +---- List<String>
    +---- ArrayList<String>
```

CODE: SRC

LetterCombinations.java

```

import java.util.*;

public class LetterCombinations {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter digits: ");
        String digits = sc.nextLine();

        List<String> result = new ArrayList<>();

        String[] phone = {
            "", "", "abc", "def", "ghi",
            "jkl", "mno", "pqrs", "tuv", "wxyz"
        };

        if (digits.length() == 0) {
            System.out.println("[]");
            return;
        }

        result.add("");

        for (int i = 0; i < digits.length(); i++) {

            int digit = digits.charAt(i) - '0';

            List<String> current = new ArrayList<>();

            for (String combination : result) {

                for (char letter : phone[digit].toCharArray()) {
                    current.add(combination + letter);
                }
            }

            result = current;
        }

        System.out.println("Letter combinations:");
    }
}

```

```
        for (String combination : result) {
            System.out.print(combination + " ");
        }

        sc.close();
    }
}
```

Compilation and Execution

```
javac LetterCombinations.java
java LetterCombinations
```

Output

```
Enter digits: 23
Letter combinations:
ad ae af bd be bf cd ce cf
```